MA1522 Linear Algebra for Computing

Lesson 22: Revision of Chapters 3 and 4

Announcements

- **Final Exam**: **Tuesday 24 November 2026, 5:00 pm**. Closed book; one helpsheet is permitted. See the **Instruction of Final Exam** file on Canvas.

- Please do the **Student Feedback Exercise**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson22` (e.g. `A1234567Z_lesson22.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 22**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Key definitions: Chapters 3 and 4

Chapter 3		
Concept	**Definition**	**Algorithm**
span(S)		$\times$
$\mathbf{v}$ is in the span of S		
S spans $\mathbb{R}^n$		
span(T) $\subseteq$ span(S)		
V is a subspace of $\mathbb{R}^n$		$\times$
The solution space of $A\mathbf{x} = \mathbf{0}$		
The solution set of $A\mathbf{x} = \mathbf{b}$ as an affine space		
S is linearly independent		
S is a basis for V		
Extend S to a basis for V		
$[\mathbf{v}]_S$, the coordinate vector of $\mathbf{v}$ relative to S		
dim(V)		
The transition matrix from T to S		

Chapter 4		
Concept	**Definition**	**Algorithm**
Row(A)		
Col(A)		
Null(A)		
rank(A)		
nullity(A)		
A is of full rank		
A is invertible		

Here S, T are subsets of $\mathbb{R}^n$, V is a subspace of $\mathbb{R}^n$, $\mathbf{v} \in \mathbb{R}^n$, and A is an $m \times n$ matrix.

Warning

1. This is not necessarily an exhaustive list.
2. You should also know their properties.

Question 1

Let a be a real number, and let

$$U = \left\{ (x, y)^T \in \mathbb{R}^2 \;\middle|\; x^2 + 2axy + y^2 = 0 \right\}.$$

1. Find all values of a for which U is a subspace of $\mathbb{R}^2$.

2. For each such value of a, find a basis for U and $\dim(U)$.

(Hint: $x^2 + 2axy + y^2 = (x + ay)^2 + (1 - a^2)\, y^2$.)

Question 1 — solution: part 1

1. Use the hint: $x^2 + 2axy + y^2 = (x + ay)^2 + (1 - a^2)\, y^2$.

   - $|a| < 1$: both terms are ≥ 0, so the sum is 0 only if $y = 0$ and $x + ay = 0$, that is, $x = y = 0$. $U = \{\mathbf{0}\}$ is a subspace.

   - $a = \pm 1$: the equation is $(x \pm y)^2 = 0$, that is, $x \pm y = 0$. U is the solution set of a homogeneous linear equation, so it is a subspace.

   - $|a| > 1$: $r^2 + 2ar + 1 = 0$ has two distinct real roots r_1, r_2, with $r_1 + r_2 = -2a$. Then $\mathbf{u} = (1, r_1)^T$ and $\mathbf{v} = (1, r_2)^T$ lie in U, but $\mathbf{u} + \mathbf{v} = (2, -2a)^T$ gives

     $$2^2 + 2a(2)(-2a) + (-2a)^2 = 4 - 4a^2 \neq 0,$$

     so $\mathbf{u} + \mathbf{v} \notin U$: U is not closed under addition, and is not a subspace.

So U is a subspace of $\mathbb{R}^2$ if and only if $-1 \leq a \leq 1$.

Question 1 — solution: part 2

2. Write the solutions of the equation in terms of free variables.

- $|a| < 1$: $U = \{\mathbf{0}\}$:

  basis $\varnothing$ (the empty set), $\qquad \dim(U) = 0.$

- $a = 1$: $x = -y$; let $y = s$: $(x, y)^T = s(-1, 1)^T$:

  $$U = \operatorname{span}\{(-1, 1)^T\}, \qquad \text{basis } \{(-1, 1)^T\}, \qquad \dim(U) = 1.$$

- $a = -1$: $x = y$; let $y = s$: $(x, y)^T = s(1, 1)^T$:

  $$U = \operatorname{span}\{(1, 1)^T\}, \qquad \text{basis } \{(1, 1)^T\}, \qquad \dim(U) = 1.$$

For $a = \pm 1$, U is a line through the origin; for $|a| < 1$, it is the origin alone. □

Question 2

Let a be a real number, and let

$$\mathbf{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \ \mathbf{s}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \ \mathbf{s}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \qquad \mathbf{t}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 3 \end{pmatrix}, \ \mathbf{t}_2 = \begin{pmatrix} 3 \\ 4 \\ 1 \\ 8 \end{pmatrix}, \ \mathbf{t}_3 = \begin{pmatrix} 3 \\ a+2 \\ 2-a \\ a+7 \end{pmatrix}.$$

Let $V = \text{span}\{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$; then $S = \{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$ is a basis for V. Let $T = \{\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3\}$.

1. Find all values of a for which $\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3$ all lie in V.

2. Find all values of a for which T is a basis for V.

3. Let $a = 1$. Find the transition matrix from S to T.

Question 2 — solution: part 1

1. Reduce $(\,\mathbf{s}_1\ \mathbf{s}_2\ \mathbf{s}_3 \mid \mathbf{t}_1\ \mathbf{t}_2\ \mathbf{t}_3\,)$ to its reduced row-echelon form: $R_3 - 2R_1$, $R_4 - R_1$; $R_3 + R_2$, $R_4 - R_2$; $R_4 + R_3$; $-R_3$; $R_1 - R_3$, $R_2 - R_3$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 2 & -1 & 0 & 0 & 1 & 2-a \\ 1 & 1 & 3 & 3 & 8 & a+7 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 0 & 0 & -1 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & 1 \\ 0 & 1 & 0 & 2 & 3 & a \\ 0 & 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

The last row is zero for every a, so each system $(\,\mathbf{s}_1\ \mathbf{s}_2\ \mathbf{s}_3 \mid \mathbf{t}_j\,)$ is consistent:

$$\mathbf{t}_1 = \mathbf{s}_1 + 2\mathbf{s}_2, \qquad \mathbf{t}_2 = 2\mathbf{s}_1 + 3\mathbf{s}_2 + \mathbf{s}_3, \qquad \mathbf{t}_3 = \mathbf{s}_1 + a\,\mathbf{s}_2 + 2\mathbf{s}_3.$$

So $\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3$ lie in V for every value of a.

Question 2 — solution: part 2

2. By part 1, $T \subseteq V$ for every a. The right block is the matrix of coordinate vectors

$$P = \begin{pmatrix} [\mathbf{t}_1]_S & [\mathbf{t}_2]_S & [\mathbf{t}_3]_S \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & a \\ 0 & 1 & 2 \end{pmatrix},$$

the transition matrix from T to S.

Since $c_1\mathbf{t}_1 + c_2\mathbf{t}_2 + c_3\mathbf{t}_3 = \begin{pmatrix} \mathbf{s}_1 & \mathbf{s}_2 & \mathbf{s}_3 \end{pmatrix} P\mathbf{c}$ and S is linearly independent, T is linearly independent if and only if the columns of P are. Reduce P: $R_2 - 2R_1$; then $R_3 + R_2$:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & a \\ 0 & 1 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & a-2 \\ 0 & 1 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & a-2 \\ 0 & 0 & a \end{pmatrix}.$$

So T is linearly independent if and only if $a \neq 0$, and then T is a set of $3 = \dim(V)$ linearly independent vectors in V: T is a basis for V if and only if $a \neq 0$.

Question 2 — solution: part 3

3. Let $a = 1$. The transition matrix from S to T is P^{-1}. Reduce $(P \mid I)$: $R_2 - 2R_1$; $R_3 + R_2$; $-R_2$; $R_2 - R_3$, $R_1 - R_3$; $R_1 - 2R_2$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -5 & 3 & 1 \\ 0 & 1 & 0 & 4 & -2 & -1 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right).$$

The transition matrix from S to T is $P^{-1} = \begin{pmatrix} -5 & 3 & 1 \\ 4 & -2 & -1 \\ -2 & 1 & 1 \end{pmatrix}$.

Check column 1, $[\mathbf{s}_1]_T = (-5, 4, -2)^T$:
$-5\mathbf{t}_1 + 4\mathbf{t}_2 - 2\mathbf{t}_3 = -5(1, 2, 0, 3)^T + 4(3, 4, 1, 8)^T - 2(3, 3, 1, 8)^T = (1, 0, 2, 1)^T = \mathbf{s}_1$. ✓ □

Question 3

Let c be a real number, and let A be a 5×4 matrix with rows $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$ such that

$$\text{rref}(A) = \begin{pmatrix} 1 & 0 & c & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \text{rref}(A^T) = \begin{pmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & c & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

1. Find all values of c for which $\{\mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}$ is a basis for $\text{Row}(A)$.

2. Find all values of c for which $\mathbf{v} = (2, 1, 4, -5)$ lies in $\text{Row}(A)$.

Question 3 — solution: part 1

1. The columns of A^T are $\mathbf{a}_1^T, \ldots, \mathbf{a}_5^T$, and $\mathrm{Row}(A) = \mathrm{Col}(A^T)$. The pivot columns of $\mathrm{rref}(A^T)$ are columns $1, 2, 4$, so $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4\}$ is a basis for $\mathrm{Row}(A)$, and $\dim \mathrm{Row}(A) = 3$.

Row operations keep the linear relations among the columns. Read the non-pivot columns 3 and 5 of $\mathrm{rref}(A^T)$:

$$\mathbf{a}_3 = 2\mathbf{a}_1 + c\,\mathbf{a}_2, \qquad \mathbf{a}_5 = -\mathbf{a}_1 + 3\mathbf{a}_4.$$

Relative to the basis $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4\}$, the coordinate vectors of $\mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$ are the columns of

$$\begin{pmatrix} 2 & 0 & -1 \\ c & 0 & 0 \\ 0 & 1 & 3 \end{pmatrix}, \qquad \det = -c \quad \text{(cofactor expansion along row 2).}$$

So $\{\mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}$ is linearly independent if and only if $c \neq 0$, and then it is a set of $3 = \dim \mathrm{Row}(A)$ linearly independent vectors in $\mathrm{Row}(A)$: a basis if and only if $c \neq 0$.

Question 3 — solution: part 2

2. Row operations do not change the row space, so $\mathrm{Row}(A) = \mathrm{Row}(\mathrm{rref}(A))$, with basis the nonzero rows

$$\mathbf{r}_1 = (1, 0, c, 0), \qquad \mathbf{r}_2 = (0, 1, -2, 0), \qquad \mathbf{r}_3 = (0, 0, 0, 1).$$

The pivot entries sit in columns $1, 2, 4$, so a combination $x_1\mathbf{r}_1 + x_2\mathbf{r}_2 + x_3\mathbf{r}_3$ has entries x_1, x_2, x_3 in columns $1, 2, 4$. If $\mathbf{v} \in \mathrm{Row}(A)$, the coefficients must be $2, 1, -5$:

$$2\mathbf{r}_1 + \mathbf{r}_2 - 5\mathbf{r}_3 = (2,\ 1,\ 2c - 2,\ -5).$$

This equals $\mathbf{v} = (2, 1, 4, -5)$ if and only if $2c - 2 = 4$:

$$\mathbf{v} \in \mathrm{Row}(A) \iff c = 3.$$

Check at $c = 3$: $2(1, 0, 3, 0) + (0, 1, -2, 0) - 5(0, 0, 0, 1) = (2, 1, 4, -5)$. ✓ □

Next lesson

Next lesson (**Fri 13 Nov**), we finish the revision with

1. Chapter 5: Orthogonality, Projection, and Least Squares Solution, and

2. Chapter 6: Eigenanalysis.

Please bring the questions you would like to go through.